# ADVANCED MOVIE SENTIMENT ANALYSIS USING NLP, MACHINE LEARNING AND DEEP LEARNING

Major Project Report

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# Abstract

This project develops an end-to-end movie review sentiment analysis system using Natural Language Processing (NLP), classical Machine Learning, recurrent neural networks, and a Transformer-based model. The study uses 50,000 labeled IMDb movie reviews, evenly divided between positive and negative classes. Exploratory data analysis was followed by TF-IDF-based classical classifiers, LSTM and BiLSTM models, and fine-tuning of DistilBERT. The evaluated classical models achieved accuracies of 89.95% for Logistic Regression, 89.11% for Linear SVM, and 86.83% for Multinomial Naive Bayes. The LSTM and BiLSTM models achieved 75.63% and 77.07%, respectively. The fine-tuned DistilBERT model achieved 91.02% accuracy on the 25,000-review test set. Error analysis identified 2,244 incorrect predictions, consisting of 1,180 false positives and 1,064 false negatives. Captum with Integrated Gradients was also explored for token-level interpretability. A Streamlit application was developed to provide interactive sentiment prediction using the exported DistilBERT model.



# Keywords

Sentiment Analysis; Natural Language Processing; IMDb; TF-IDF; LSTM; BiLSTM; DistilBERT; Transformer; Captum; Streamlit


# 1. Introduction

Sentiment analysis is an NLP task that identifies the polarity expressed in textual data. For movie reviews, the task can be formulated as binary classification into positive and negative sentiment. This project extends a conventional TF-IDF and machine-learning approach into a comparative study involving recurrent neural networks and a pretrained Transformer.

The project follows an end-to-end workflow covering dataset exploration, cleaning, feature representation, model training, evaluation, error analysis, explainability, model export, and interactive deployment.

# 2. Problem Statement

Movie reviews contain subjective language, negation, context, and long-range relationships between words. The project investigates how performance changes as the representation and model family evolve from TF-IDF classifiers to recurrent neural networks and a pretrained Transformer.

# 3. Objectives

• Analyze and clean a large labeled movie-review dataset.

• Perform exploratory data analysis.

• Build TF-IDF-based classical Machine Learning baselines.

• Implement LSTM and BiLSTM models.

• Fine-tune DistilBERT for binary sentiment classification.

• Compare models on a common held-out test set.

• Perform error analysis and explore Integrated Gradients.

• Deploy the final model through Streamlit.

# 4. Dataset and Exploratory Data Analysis

The dataset contains 50,000 labeled IMDb reviews: 25,000 training and 25,000 testing reviews, balanced between positive and negative sentiment. The training split was divided into 20,000 training and 5,000 validation examples for DistilBERT. The original 25,000-review test set was retained for final evaluation.

EDA found no missing values. The analysis also detected 96 duplicate reviews in training, 199 duplicate reviews in testing, and 123 review texts appearing in both train and test. These are properties of the provided dataset copy.

Training review lengths had a mean of approximately 233.8 words, median of 174, and maximum of 2,470 words. The 90th, 95th and 99th percentiles were approximately 458, 598 and 913 words.

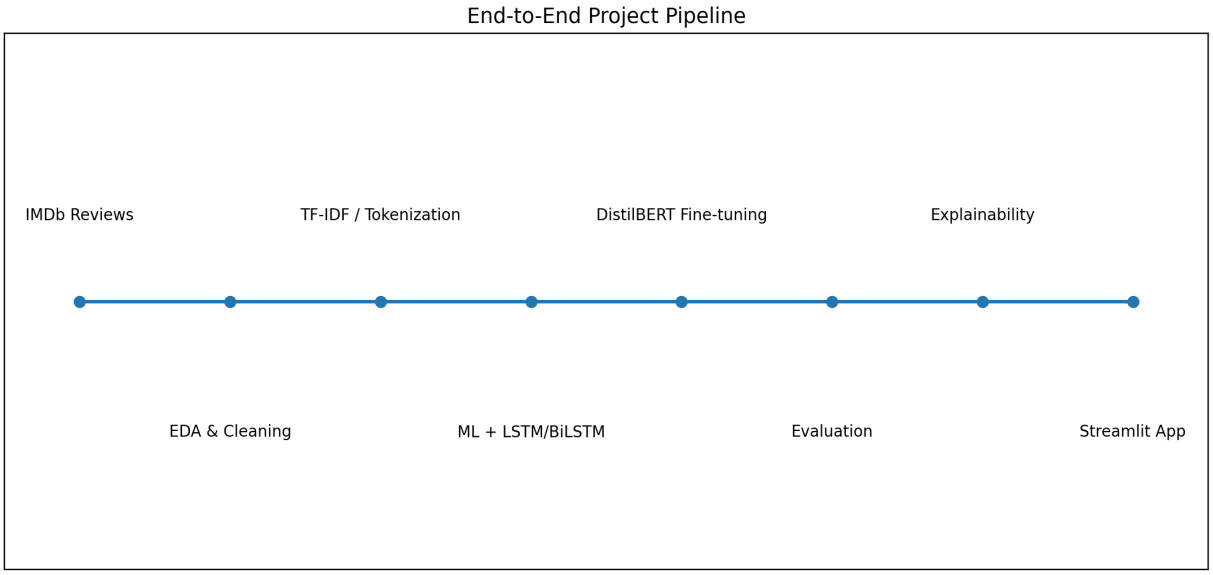



Figure 1. End-to-end project workflow.

# 5. Methodology

The workflow includes dataset validation, exploratory analysis, HTML/text cleaning, TF-IDF feature extraction, sequence preparation for recurrent models, DistilBERT tokenization, model training, final test evaluation, error analysis, explainability, and Streamlit deployment.

# 6. Classical Machine Learning Models

TF-IDF transformed the reviews into numerical representations. The resulting training matrix contained 25,000 reviews and 20,000 features. Logistic Regression achieved 89.95%, Linear SVM 89.11%, and Multinomial Naive Bayes 86.83%.

# 7. Deep Learning Models

## 7.1 LSTM

The CPU-friendly LSTM used an embedding layer, LSTM layer, dropout, and fully connected output. It achieved 75.63% test accuracy.

## 7.2 BiLSTM

The BiLSTM processed sequences in both directions and achieved 77.07% test accuracy.

# 8. Transformer-Based Sentiment Analysis

DistilBERT was selected as a compact pretrained Transformer. Fine-tuning used distilbert-base-uncased, a maximum sequence length of 256, learning rate 2e-5, batch sizes of 16/32, three epochs, weight decay 0.01, and FP16 on an NVIDIA Tesla T4 GPU.

Validation accuracy reached 90.72%, 91.50%, and 91.96% across epochs 1–3. Final test accuracy was 91.024%.

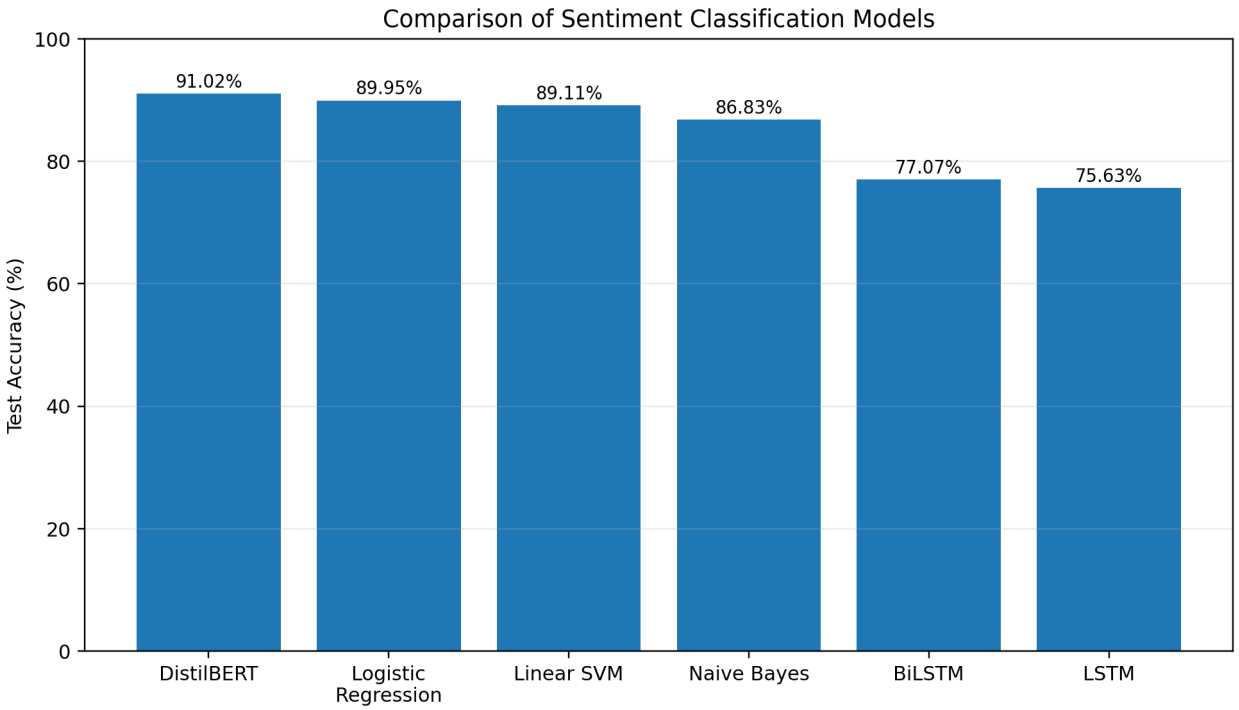



Figure 2. Test accuracy comparison.

## 9. Explainable AI

Captum with Integrated Gradients was used to estimate token-level contribution to a Transformer prediction. Special tokens were excluded from user-facing interpretation.

## 10. Error Analysis

DistilBERT produced 22,756 correct predictions and 2,244 errors on the 25,000-review test set. There were 1,180 false positives and 1,064 false negatives.

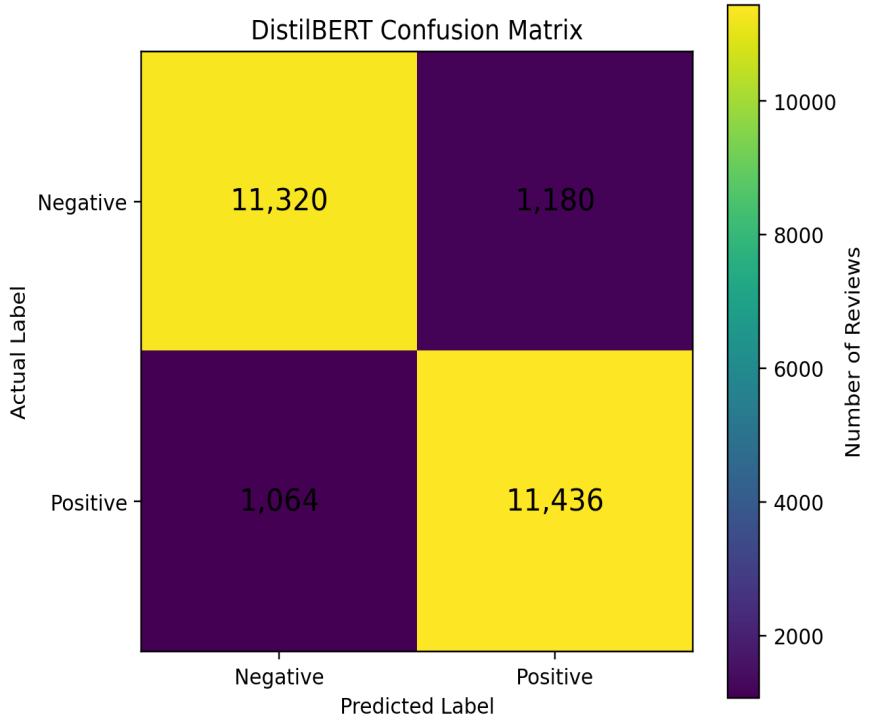



Figure 3. DistilBERT confusion matrix.

## 11. Streamlit Application

A Streamlit application accepts a movie review and returns the predicted sentiment and confidence. It also displays model information and a comparison chart. The exported DistilBERT model and tokenizer were successfully loaded locally and tested with positive and negative review examples.

# 12. Results and Discussion

| Model | Test Accuracy |
|---|---|
| DistilBERT | 91.02% |
| Logistic Regression | 89.95% |
| Linear SVM | 89.11% |
| Naive Bayes | 86.83% |
| BiLSTM | 77.07% |
| LSTM | 75.63% |

DistilBERT achieved 91.02% on the evaluated test set, followed by Logistic Regression at 89.95% and Linear SVM at 89.11%. The recurrent models were evaluated under a computationally constrained setup and produced lower measured accuracies. Validation loss for DistilBERT increased after epoch 1 while validation accuracy continued to rise, indicating a possible overfitting trend.

# 13. Limitations

• Observed duplicate and cross-split review texts are present in the provided dataset.

• LSTM and BiLSTM used reduced training subsets for computational efficiency.

• DistilBERT truncates inputs beyond 256 tokens.

• Sarcasm, mixed sentiment, negation, and subtle context remain challenging.

• Confidence is not a guarantee of correctness.

# 14. Future Scope

• Additional labeled/unlabeled data and broader evaluation.

• Alternative and larger Transformer architectures.

• Systematic hyperparameter optimization.

• More detailed error categorization.

• Richer explainability visualization.

• Cloud deployment and multilingual/emotion classification.

# 15. Conclusion

This project developed a complete movie sentiment analysis pipeline spanning classical NLP, Machine Learning, recurrent Deep Learning, Transformer fine-tuning, explainability, and deployment. The evaluated accuracies were 89.95% for Logistic Regression, 89.11% for Linear SVM, 86.83% for Naive Bayes, 75.63% for LSTM, 77.07% for BiLSTM, and 91.02% for DistilBERT. The resulting Streamlit application demonstrates how the fine-tuned Transformer can be used in an interactive sentiment analysis system.

# 16. References

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Project experimental notebooks, trained models, evaluation outputs, and Streamlit implementation.